

Nearly Spherical Closure for Route A: sharp Saint–Venant stability via BDV Theorem 3.3 and Lemma 4.2

Plan 1 / Route A — block `NearlySphericalClosure`

Abstract

We record, as a self-contained pointwise quantitative statement, the “nearly spherical closure” building block of Route A. Starting from two quantitative results in Brasco–De Philippis–Velichkov [1] (Theorem 3.3 and Lemma 4.2), we produce, with explicit dimensional constants, the one-step inequality

$$E(\tilde{\Omega}) - E(B_1) \geq c_*(N) \alpha(\tilde{\Omega}) \quad \text{and} \quad E(\tilde{\Omega}) - E(B_1) \geq c_*(N) |B_1|^2 C_1(N)^{-1} \mathcal{A}(\tilde{\Omega})^2,$$

valid for every C^{2,γ_0} nearly spherical set $\tilde{\Omega}$ with sufficiently small parametrisation. Here E denotes the Saint–Venant (torsion) energy used by BDV; the transfer to the first Dirichlet eigenvalue is performed downstream by the `KohlerJobinTransfer` block. The whole argument is by direct substitution into the BDV estimates: no compactness, no contradiction and no selection principle are used here.

1 Setup, conventions and the BDV estimates

Throughout, $N \geq 2$ and $0 < \gamma_0 \leq 1$ are fixed. We write $|\cdot| = \mathcal{L}^N$ for Lebesgue measure and $\mathcal{H}^{N-1}$ for the surface measure on ∂B_1 .

Saint–Venant (torsion) energy. Following [1, §2], for every open bounded set $\Omega \subset \mathbb{R}^N$ we denote by $u_\Omega \in W_0^{1,2}(\Omega)$ the unique weak solution of

$$-\Delta u_\Omega = 1 \quad \text{in } \Omega, \quad u_\Omega = 0 \quad \text{on } \partial\Omega,$$

and by

$$T(\Omega) := \int_{\Omega} u_\Omega \, dx$$

the torsional rigidity. The Saint–Venant energy is

$$E(\Omega) := -\frac{1}{2} T(\Omega) = \inf_{v \in W_0^{1,2}(\Omega)} \left(\frac{1}{2} \int_{\Omega} |\nabla v|^2 \, dx - \int_{\Omega} v \, dx \right). \quad (1)$$

This is the energy stabilised by Theorem 3.3 of [1]. The transfer to the first Dirichlet eigenvalue $\lambda_1(\Omega) = \inf\{\int |\nabla u|^2 : u \in W_0^{1,2}(\Omega), \|u\|_{L^2} = 1\}$ is performed by the Köhler–Jobin inequality and is the subject of a separate block (`KohlerJobinTransfer`).

Fraenkel and smooth asymmetries. The (classical) *Fraenkel asymmetry* is

$$\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^N} \frac{|\Omega \Delta B_1(x_0)|}{|B_1|}, \quad |\Omega| = |B_1|.$$

Following [1, Definition 4.1] we also introduce the *smooth asymmetry*

$$\alpha(\Omega) := \int_{\Omega \Delta B_1(x_\Omega)} |1 - |x - x_\Omega|| dx, \quad x_\Omega = \frac{1}{|\Omega|} \int_{\Omega} x dx. \quad (2)$$

The smooth asymmetry α weights the symmetric difference by the distance to the unit sphere, while $\mathcal{A}$ uses the unweighted symmetric difference. The inequality between them is Lemma 1.2(i) below; it goes in the direction $\mathcal{A}^2 \lesssim \alpha$.

Nearly spherical sets. Following [1, Definition 3.1] we say that an open bounded set $\tilde{\Omega} \subset \mathbb{R}^N$ is *nearly spherical of class C^{2,γ_0} with parametrisation g* if there exists $g \in C^{2,\gamma_0}(\partial B_1)$ with $\|g\|_{L^\infty(\partial B_1)} \leq \frac{1}{2}$ such that

$$\partial \tilde{\Omega} = \{(1 + g(y))y : y \in \partial B_1\}, \quad \tilde{\Omega} = \{ry : y \in \partial B_1, 0 \leq r < 1 + g(y)\}.$$

The $H^{1/2}(\partial B_1)$ norm is, as in [1, Definition 3.2],

$$\|g\|_{H^{1/2}(\partial B_1)}^2 := \int_{\partial B_1} g^2 d\mathcal{H}^{N-1} + \int_{B_1} |\nabla H(g)|^2 dx,$$

where $H(g)$ is the $W^{1,2}$ -harmonic extension of g to B_1 .

The two quantitative statements we use as a black box are now recalled.

Lemma 1.1 (BDV Theorem 3.3, sharp nearly spherical stability). *For every $0 < \gamma_0 \leq 1$ there exists $\delta_1 = \delta_1(N, \gamma_0) > 0$ such that, for every nearly spherical set Ω of class C^{2,γ_0} parametrised by g with*

$$\|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_1, \quad |\Omega| = |B_1|, \quad x_\Omega = 0,$$

one has

$$E(\Omega) - E(B_1) \geq \frac{1}{32N^2} \|g\|_{H^{1/2}(\partial B_1)}^2, \quad (3)$$

with E the Saint–Venant energy (1).

The dimensional constant $c_{\text{sph}}(N) := \frac{1}{32N^2}$ is the value of the bilinear-form coefficient appearing in [1, (3.9)–(3.10)]; it comes from the gap to the second Steklov eigenvalue of B_1 (equal to 2 in every dimension).

Lemma 1.2 (BDV Lemma 4.2). *There exist constants $C_1 = C_1(N) > 0$, $\delta_3 = \delta_3(N) > 0$ and $C_3 = C_3(N) > 0$ with the following properties.*

(i) *For every measurable set $\Omega \subset \mathbb{R}^N$ with $|\Omega| = |B_1|$,*

$$C_1(N) \alpha(\Omega) \geq |\Omega \Delta B_1(x_\Omega)|^2 \geq |B_1|^2 \mathcal{A}(\Omega)^2. \quad (4)$$

(iii) *For every nearly spherical set Ω parametrised by g with $\|g\|_{L^\infty(\partial B_1)} \leq \delta_3$,*

$$\alpha(\Omega) \leq C_3(N) \|g\|_{L^2(\partial B_1)}^2. \quad (5)$$

Reference. Items (i) and (iii) are [1, Lemma 4.2 (i),(iii)]; the proof there is by direct rearrangement / Taylor expansion of the weighted integral (2). We use only these two items in the present section; in particular we do *not* use the BDV selection principle nor Lemma 4.2(ii). $\square$

2 The nearly spherical closure

2.1 Statement in the volume- and barycenter-normalised case

The trivial trace–Poincaré comparison

$$\|g\|_{L^2(\partial B_1)}^2 \leq \|g\|_{H^{1/2}(\partial B_1)}^2, \quad (6)$$

follows directly from the definition of $\|\cdot\|_{H^{1/2}(\partial B_1)}$ (L^2 is one of the two summands).

Theorem 2.1 (NearlySphericalClosure: nearly spherical closure). *Fix $0 < \gamma_0 \leq 1$ and set*

$$\delta_{\text{sph}}(N, \gamma_0) := \min(\delta_1(N, \gamma_0), \delta_3(N)). \quad (7)$$

For every C^{2,γ_0} nearly spherical set $\tilde{\Omega} = \{(1 + g(y))y : y \in \partial B_1\}$ with

$$\|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}(N, \gamma_0), \quad |\tilde{\Omega}| = |B_1|, \quad x_{\tilde{\Omega}} = 0, \quad (8)$$

we have

$$E(\tilde{\Omega}) - E(B_1) \geq \frac{c_{\text{sph}}(N)}{C_3(N)} \alpha(\tilde{\Omega}) =: c_*(N) \alpha(\tilde{\Omega}), \quad (9)$$

$$E(\tilde{\Omega}) - E(B_1) \geq \frac{c_{\text{sph}}(N) |B_1|^2}{C_3(N) C_1(N)} \mathcal{A}(\tilde{\Omega})^2 = c_*(N) |B_1|^2 C_1(N)^{-1} \mathcal{A}(\tilde{\Omega})^2, \quad (10)$$

with the explicit dimensional constant

$$c_*(N) = \frac{c_{\text{sph}}(N)}{C_3(N)} = \frac{1}{32 N^2 C_3(N)}. \quad (11)$$

Equivalently, the smooth-asymmetry energy quotient is bounded below by an explicit dimensional constant:

$$Q_\alpha(\tilde{\Omega}) := \frac{E(\tilde{\Omega}) - E(B_1)}{\alpha(\tilde{\Omega})} \geq c_*(N). \quad (12)$$

Proof. By (7), the hypotheses of Lemma 1.1 (threshold δ_1) and of Lemma 1.2(iii) (threshold δ_3) hold simultaneously.

Step 1: smooth-asymmetry form (9). Chain Lemma 1.1, the comparison (6), and Lemma 1.2(iii):

$$E(\tilde{\Omega}) - E(B_1) \stackrel{(3)}{\geq} \frac{1}{32 N^2} \|g\|_{H^{1/2}(\partial B_1)}^2 \stackrel{(6)}{\geq} \frac{1}{32 N^2} \|g\|_{L^2(\partial B_1)}^2 \stackrel{(5)}{\geq} \frac{1}{32 N^2 C_3(N)} \alpha(\tilde{\Omega}).$$

This is (9).

Step 2: Fraenkel form (10). Combining with Lemma 1.2(i):

$$E(\tilde{\Omega}) - E(B_1) \geq c_*(N) \alpha(\tilde{\Omega}) \stackrel{(4)}{\geq} \frac{c_*(N)}{C_1(N)} |B_1|^2 \mathcal{A}(\tilde{\Omega})^2.$$

The quotient identity (12) is (9) rewritten. □

Remark 2.2 (Direction of asymmetry inequalities). The inequalities (4) and (5) go in opposite directions: α is *lower bounded* by $\mathcal{A}^2$ (so α is the larger, easier-to-bound-below quantity) and *upper bounded* by $\|g\|_{L^2(\partial B_1)}^2$. Theorem 2.1 uses both in the favourable direction: (5) converts the $H^{1/2}$ energy lower bound into an α -lower bound, while (4) transfers it to $\mathcal{A}^2$.

2.2 Absorbing the volume/barycenter normalisation

In Route A the set $\tilde{\Omega}$ produced upstream is C^{2,γ_0} -close to B_1 but may have $|\tilde{\Omega}| \neq |B_1|$ or $x_{\tilde{\Omega}} \neq 0$. The normalisation is contained in BDV's own proof of Theorem 3.3 (see [1, (3.6)–(3.7)]): expanding the two scalar identities

$$|\tilde{\Omega}| = \int_{\partial B_1} \frac{(1+g)^N}{N} d\mathcal{H}^{N-1} = |B_1|, \quad (13)$$

$$x_{\tilde{\Omega}} = \int_{\partial B_1} y \frac{(1+g)^{N+1}}{N+1} d\mathcal{H}^{N-1} = 0, \quad (14)$$

in g and using $\|g\|_{C^{2,\gamma_0}} \leq \delta_1$ yields the moment bounds [1, (3.6)–(3.7)]

$$\left| \int_{\partial B_1} g d\mathcal{H}^{N-1} \right| + \sum_{i=1}^N \left| \int_{\partial B_1} y_i g d\mathcal{H}^{N-1} \right| \leq C(N) \delta_1 \|g\|_{H^{1/2}(\partial B_1)}. \quad (15)$$

Corollary 2.3 (Normalisation absorbed in the constant). *If $\tilde{\Omega}$ satisfies all hypotheses of Theorem 2.1 except that $|\tilde{\Omega}| = |B_1|$ and $x_{\tilde{\Omega}} = 0$ are replaced by*

$$||\tilde{\Omega}| - |B_1|| + |x_{\tilde{\Omega}}| \leq \eta \|g\|_{L^2(\partial B_1)},$$

with $\eta \leq \eta_0(N, \gamma_0)$ small enough, then there exist $\tilde{c}_(N) > 0$ and $\tilde{\delta}_{\text{sph}}(N, \gamma_0) > 0$ such that conclusions (9)–(10) hold with $c_*(N)$ replaced by $\tilde{c}_*(N)$. Equivalently, the volume–barycenter normalisation in (8) can be replaced by an $O(\|g\|_{L^2(\partial B_1)})$ defect at the cost of a dimensional constant.*

Reduction. Form $\tilde{\Omega}' = \tilde{\Omega} - x_{\tilde{\Omega}}$ and $\tilde{\Omega}'' = \lambda \tilde{\Omega}'$ with $\lambda = (|B_1|/|\tilde{\Omega}'|)^{1/N}$. Then $|\lambda - 1| \leq C\eta \|g\|_{L^2}$ and $|x_{\tilde{\Omega}''}| \leq \eta \|g\|_{L^2}$. The new parametrisation g'' of $\tilde{\Omega}''$ satisfies $\|g''\|_{C^{2,\gamma_0}} \leq \|g\|_{C^{2,\gamma_0}} + C(N)\eta \|g\|_{L^2} \leq 2\|g\|_{C^{2,\gamma_0}}$ for η small. By construction $|\tilde{\Omega}''| = |B_1|$ and $x_{\tilde{\Omega}''} = 0$, and the perturbations of $E, \alpha, \mathcal{A}$ are $O(\eta \|g\|_{L^2}^2)$, absorbed at the cost of a factor $\frac{1}{2}$ in the conclusion. Setting $\tilde{c}_*(N) = c_*(N)/4$ closes the argument. $\square$

2.3 Sharpness

The exponent 2 on $\mathcal{A}$ in (10) (equivalently the exponent 1/2 on the deficit in the rearranged inequality $\mathcal{A}(\tilde{\Omega}) \leq C(E(\tilde{\Omega}) - E(B_1))^{1/2}$) cannot be improved, already inside the nearly spherical class. This is shown in [1, p. 2 and the discussion preceding §2] by an ellipsoidal example.

Remark 2.4 (Ellipsoidal sharpness). Let $E_\varepsilon = \{x : \sum x_i^2/a_i^2 < 1\}$ with $a_i = 1 + \varepsilon e_i$, $\sum e_i = 0$, $|e| \leq 1$. A direct expansion gives

$$\mathcal{A}(E_\varepsilon) = c_1(N, e) \varepsilon + O(\varepsilon^2), \quad E(E_\varepsilon) - E(B_1) = c_2(N, e) \varepsilon^2 + O(\varepsilon^3),$$

so $\mathcal{A}^2$ and the energy deficit are of the same order ε^2 . No inequality $E(\tilde{\Omega}) - E(B_1) \geq c \mathcal{A}(\tilde{\Omega})^{2-\theta}$ ($\theta > 0$) can hold uniformly even within nearly spherical sets, so the exponent 2 on $\mathcal{A}$ (equivalently 1/2 on the deficit) is sharp.

3 Summary of constants

Symbol	Definition / origin	Where
$c_{\text{sph}}(N)$	$\frac{1}{32 N^2}$, coefficient in BDV Thm 3.3	(3)
$\delta_1(N, \gamma_0)$	C^{2, γ_0} smallness in BDV Thm 3.3	Lem. 1.1
$C_1(N)$	$C_1 \alpha \geq \Omega \triangle B_1(x_\Omega) ^2$, BDV 4.2(i)	(4)
$\delta_3(N)$	L^∞ smallness in BDV Lem. 4.2(iii)	(5)
$C_3(N)$	$\alpha \leq C_3 \ g\ _{L^2(\partial B_1)}^2$, BDV Lem. 4.2(iii)	(5)
$\delta_{\text{sph}}(N, \gamma_0)$	$\min(\delta_1, \delta_3)$	(7)
$c_*(N)$	$\frac{1}{32 N^2 C_3(N)}$	(11)
$c_*(N) B_1 ^2 / C_1(N)$	Fraenkel constant in (10)	(10)
$\tilde{c}_*(N)$	$c_*(N)/4$ after vol./barycenter normalisation	Cor. 2.3

Pure N -dependence. All constants depend only on N , except δ_1 and hence δ_{sph} which depend additionally on γ_0 . No radius parameter R enters, since by the time the closure is invoked the upstream selection has returned a set whose convex hull is contained in a ball of dimensional radius and whose volume is normalised to $|B_1|$.

4 Use in Route A

In the Plan 1 / Route A architecture, NearlySphericalClosure (Theorem 2.1) is applied as follows.

- (R1) The upstream blocks (**SelectionPrinciple**, **GraphEntry**, **SchauderInterpolation**) produce a set $\tilde{\Omega}$ which is C^{2, γ_0} a graph over ∂B_1 with parametrisation g satisfying $\|g\|_{C^{2, \gamma_0}} \leq h_{\text{sph}}(N, \gamma_0, R)$ for some $h_{\text{sph}} > 0$.
- (R2) After volume normalisation and barycenter shift (Corollary 2.3) we may assume $|\tilde{\Omega}| = |B_1|$, $x_{\tilde{\Omega}} = 0$, and $h_{\text{sph}} \leq \delta_{\text{sph}}(N, \gamma_0)$.
- (R3) Theorem 2.1 then gives, with the explicit constants of (11),

$$E(\tilde{\Omega}) - E(B_1) \geq \frac{|B_1|^2}{32 N^2 C_3(N) C_1(N)} \mathcal{A}(\tilde{\Omega})^2,$$

the closing inequality of the sharp Saint–Venant stability with the sharp exponent 2 on $\mathcal{A}$ (equivalently 1/2 on the deficit).

What we do *not* use. The proof of Theorem 2.1 does not use:

- BDV’s selection principle ([1, Prop. 4.4]);
- BDV’s compactness/contradiction step ([1, Thm 4.3]);
- Lemma 4.2(ii) (Lipschitz dependence of α on symmetric difference).

The two BDV ingredients we do use (Theorem 3.3 and Lemma 4.2(i),(iii)) are both proved in [1] by direct spherical-harmonic and rearrangement computations.

References

- [1] L. Brasco, G. De Philippis, B. Velichkov. *Faber–Krahn inequalities in sharp quantitative form*. arXiv:1306.0392v1 (2013).